The story/background

Setting: SeeWeeS Specialty distribution in New Jersey ships time-critical medicines to Boston-area hospitals (e.g., Mass General / Brigham / etc.). Deliveries travel the I-95 corridor and must land within tight SLA windows (cold chain, oncology, ICU meds). If weather degrades, *travel time variability explodes*, and dispatch needs to re-plan *before trucks leave the dock*.

Business pain:

1. Bad weather causes late deliveries → missed appointments / OR schedules/penalties.

2. Item master data mismatches (item_id ↔ item_name) cause picking/shipping errors → returns, delays, patient risk.

3. Incomplete identifiers (missing `unique_item_id`) break traceability and packing logic → you can't reliably compute truck count.

Objective: Build a **multi-agent "Dispatch + Data Quality + Report"** that:

- Reads a PDF "Operations Playbook + Item Master Appendix."

- Audits a CSV "shipment line items."

- Pulls weather forecasts for the next 24 hours along I-95

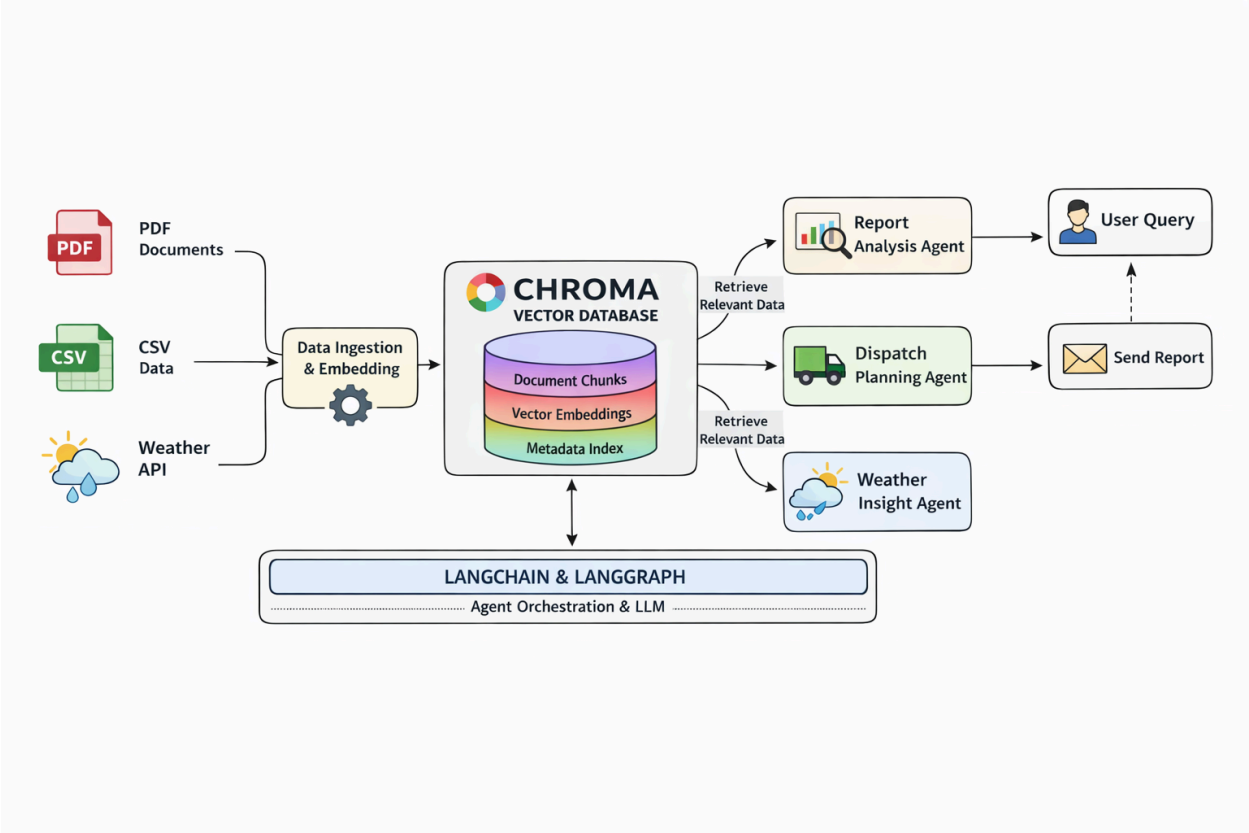
- Produces an executive-ready report (and optionally emails it)

Architecture (multi-agent, tool-using)

Agents (roles)

- **ContextAgent (PDF RAG):** extracts "business rules", KPI definitions, constraints, and dispatch heuristics from the PDF via retrieval.

- **OpsDataAgent (CSV Analyzer):** computes KPIs, checks data quality, and finds anomalies.

- **WeatherAgent:** fetches forecast + derives operational risk features (snow, heavy rain, wind gusts, temp).

- **PlannerAgent:** combines PDF context + CSV findings + weather risk into dispatch recommendations.
- **ReportAgent:** turns everything into a structured report (Markdown/HTML).
- **EmailAgent:** sends the report to an email recipient.